Interacting Multipoles of Rank 2 and 2

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^{B}_{\alpha\beta}\Theta^{A}_{\gamma\delta}=R_z^{-9}\cdot\frac{1}{9}\cdot\begin{bmatrix} +105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\delta\left(\gamma,\delta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\delta\left(\beta,\delta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\delta\left(\beta,\gamma\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\delta\left(\alpha,\delta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\delta\left(\alpha,\gamma\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\alpha,\beta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\delta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\delta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\gamma\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta} \end{bmatrix}\tag{1}$$

Simplify deltas:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^{B}_{\alpha\beta}\Theta^{A}_{\gamma\delta}=R_z^{-9}\cdot\frac{1}{9}\cdot\begin{bmatrix} +105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\delta\left(\gamma,\gamma\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\delta\left(\beta,\beta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\delta\left(\beta,\beta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\delta}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\delta\left(\alpha,\alpha\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\gamma}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\delta\left(\alpha,\alpha\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\delta}\\ -15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\alpha,\alpha\right)\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\gamma\delta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta}\\ +3\cdot R_z^4\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{bmatrix}\tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9}\cdot\frac{1}{9}\cdot\begin{bmatrix} +105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\delta}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\gamma}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\delta}\\ -15\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\gamma\delta}\\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+3\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta}\\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{bmatrix}\tag{2b}$$

$$\xRightarrow{\text{reduced indices}} R_z^{-9}\cdot\frac{1}{9}\cdot\begin{bmatrix} +105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\gamma}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\gamma}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\gamma}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\gamma}\\ -15\cdot R_z^2\cdot R_\gamma\cdot R_\beta\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\gamma}\\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+3\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta}\\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{bmatrix}\tag{2c}$$

$$\xRightarrow{\text{added up}} R_z^{-9}\cdot\frac{1}{9}\cdot\begin{bmatrix} +105\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\delta}\\ -15\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\gamma\gamma}\\ -30\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\beta\gamma}\\ -30\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\gamma}\\ -15\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\gamma}\\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+6\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{bmatrix}\tag{2d}$$

Simplify R:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^B_{\alpha\beta}\Theta^A_{\gamma\delta} = R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +105\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ -15\cdot R_z^2\cdot R_z\cdot R_z\cdot\Theta^B_{zz}\cdot\Theta^A_{\gamma\gamma} \\ -30\cdot R_z^2\cdot R_z\cdot R_z\cdot\Theta^B_{z\beta}\cdot\Theta^A_{z\beta} \\ -30\cdot R_z^2\cdot R_z\cdot R_z\cdot\Theta^B_{z\alpha}\cdot\Theta^A_{z\alpha} \\ -15\cdot R_z^2\cdot R_z\cdot R_z\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz} \\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+6\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +105\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz}-15\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{\gamma\gamma} \\ -30\cdot R_z^4\cdot\Theta^B_{z\beta}\cdot\Theta^A_{z\beta}-30\cdot R_z^4\cdot\Theta^B_{z\alpha}\cdot\Theta^A_{z\alpha} \\ -15\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz}+3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta} \\ +6\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{3b}$$

$$\xRightarrow{\text{reduced indices}} R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +105\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz}-15\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{\alpha\alpha} \\ -30\cdot R_z^4\cdot\Theta^B_{z\alpha}\cdot\Theta^A_{z\alpha}-30\cdot R_z^4\cdot\Theta^B_{z\alpha}\cdot\Theta^A_{z\alpha} \\ -15\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz}+3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta} \\ +6\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{3c}$$

$$\xRightarrow{\text{added up}} R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +105\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz}-15\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{\alpha\alpha} \\ -60\cdot R_z^4\cdot\Theta^B_{z\alpha}\cdot\Theta^A_{z\alpha}-15\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz} \\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+6\cdot R_z^4\cdot\Theta^B_{\alpha\beta}\cdot\Theta^A_{\alpha\beta} \end{array}\right] \tag{3d}$$

Simplify Quadrupoles:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^B_{\alpha\beta}\Theta^A_{\gamma\delta} = R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +105\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz}-15\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{\alpha\alpha} \\ -60\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz}-15\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz} \\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta}+6\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\alpha\alpha} \end{array}\right] \tag{4a}$$

$$\xRightarrow{\text{added up}} R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +45\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ -15\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{\alpha\alpha} \\ -15\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{zz} \\ +3\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\beta\beta} \\ +6\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\alpha\alpha} \end{array}\right] \tag{4b}$$

Exploit Traceless Conditions:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^B_{\alpha\beta}\Theta^A_{\gamma\delta} = R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +45\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ +6\cdot R_z^4\cdot\Theta^B_{\alpha\alpha}\cdot\Theta^A_{\alpha\alpha} \end{array}\right] \tag{5}$$

Multiply Out Sum:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^B_{\alpha\beta}\Theta^A_{\gamma\delta} = R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +45\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ +6\cdot R_z^4\cdot\Theta^B_{xx}\cdot\Theta^A_{xx} \\ +6\cdot R_z^4\cdot\Theta^B_{yy}\cdot\Theta^A_{yy} \\ +6\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \end{array}\right] \tag{6a}$$

$$\xRightarrow{\text{added up}} R_z^{-9}\cdot\frac{1}{9}\cdot\left[\begin{array}{c} +51\cdot R_z^4\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ +6\cdot R_z^4\cdot\Theta^B_{xx}\cdot\Theta^A_{xx} \\ +6\cdot R_z^4\cdot\Theta^B_{yy}\cdot\Theta^A_{yy} \end{array}\right] \tag{6b}$$

Combining everything:

$$+\frac{1}{9}\cdot T_{\alpha\beta\gamma\delta}\Theta^B_{\alpha\beta}\Theta^A_{\gamma\delta} = \left[\begin{array}{c} +\frac{17}{3}\cdot R_z^{-5}\cdot\Theta^B_{zz}\cdot\Theta^A_{zz} \\ +\frac{2}{3}\cdot R_z^{-5}\cdot\Theta^B_{xx}\cdot\Theta^A_{xx} \\ +\frac{2}{3}\cdot R_z^{-5}\cdot\Theta^B_{yy}\cdot\Theta^A_{yy} \end{array}\right] \tag{7}$$